



**STATUS** Built — working MVP, demo-ready (live + offline sample gallery)

## 1 · User Problem

Saudi regulatory and business documents — contracts, circulars, MoUs, board minutes — still live as paper, scans, photos, and handwritten notes, much of it in Arabic. Turning them into trusted, structured data is slow and manual.

- **Locked in unstructured formats** — critical fields and clauses sit inside PDFs, images, and handwriting rather than in data systems.
- **Arabic breaks generic tools** — RTL text, ligatures, handwriting, and bilingual Arabic/English layouts defeat mainstream OCR and off-the-shelf AI.
- **No audit trail** — teams retype and re-read by hand; errors slip through, and there is no link from extracted data back to the source.

## 2 · Target Audience

- **Compliance & legal teams** — tracking obligations, deadlines, parties, and clauses across regulatory documents.
- **Government-facing organizations** — digitizing large backlogs of regulatory and administrative archives.
- **Operations / back-offices** — converting paper records into structured, searchable data.
- **Primary user (persona)** — an operator/reviewer who needs a fast, trustworthy first pass they can verify and approve before archiving.

## 3 · Proposed AI Solution

**Naskh** — a human-in-the-loop document intelligence assistant. AI does an honest first pass; a person reviews, edits, and approves. AI is used in three concrete ways:

- **Vision extraction** — server-side OpenAI gpt-4o with Pydantic structured outputs returns typed fields, per-field confidence, and full Arabic transcription.
- **Grounded, cited Q&A** — a per-document ChromaDB index powers RAG so the assistant answers from the document (streaming), not generic chat.
- **Source highlighting** — fields and answers map to exact text-layer coordinates, lighting up the real region on the page for verification.

Designed Arabic-first, cited and auditable, with keys kept server-side. Honest by design: it does not claim perfect OCR.



## 4 · MVP Scope

### IN SCOPE (built)

- Upload PDF / image (PNG, JPEG, WebP)
- Faithful preview — pypdfium2 page rasterization
- Structured field extraction + confidence + review flags
- Editable Arabic transcription (RTL)
- Split-view with live source highlighting
- Cited assistant — streaming RAG answers
- Export to DOCX · JSON · CSV
- Offline sample gallery · dark / light UI

### OUT OF SCOPE (MVP)

- Accounts, authentication, multi-user
- Government / ECM system integrations
- Broad document-type coverage
- Fine-tuned Arabic handwriting model
- Persistence / database, batch processing
- Production deploy, scaling infrastructure

**Success criteria.** A real Saudi document processes end-to-end; fields extract with confidence and review flags; the assistant answers with accurate, clickable citations; and exports open cleanly — all demonstrable live, with an offline sample path as backup.